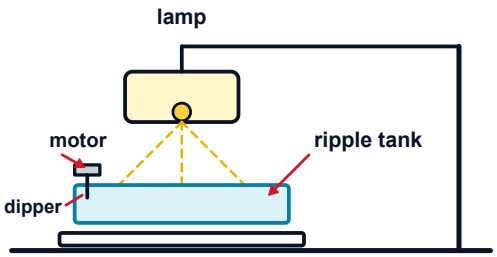
DO NOW · REQUIRED PRACTICAL

Waves in a ripple tank

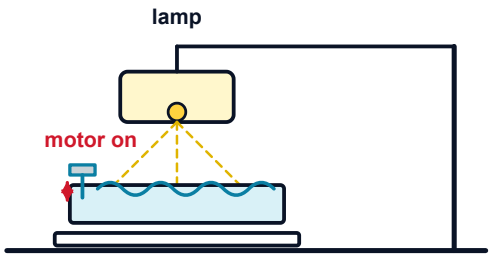
Waves
Paper 2 · Combined and Separate
6 steps

A student measures the speed of the waves in a ripple tank.

1 Set up the ripple tank with a lamp above it and a white screen below.

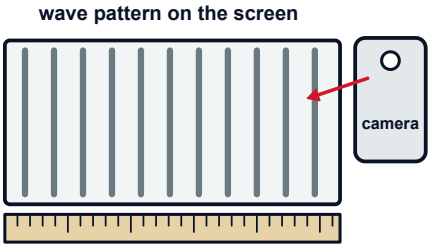


2 Switch on the motor so that the dipper makes steady waves.

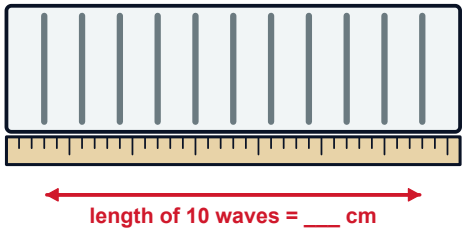


the dipper moves up and down, making steady waves

3 Place a ruler on the screen and photograph the wave pattern.

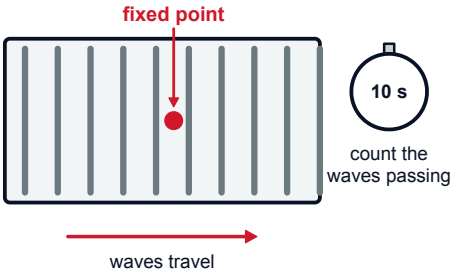


4 Measure the length of 10 waves on the photograph and divide by 10 to find the wavelength.



wavelength λ = length ÷ 10

5 Count the waves passing a point in 10 seconds and divide by 10 to find the frequency.



frequency f = number of waves ÷ 10

6 Calculate the wave speed: frequency × wavelength.

v = f × λ

wave speed = frequency × wavelength

m/s = Hz × m

use f from step 5 and λ from step 4

Note: The frequency can be found before or after the wavelength.

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